

Q. 2 (1)
A. 2 (1)

Base Quantities

Derived Quantities

Definition

① Physical quantities from which other physical quantities can be derived are known as base quantities.

② Physical quantities which can be obtained multiplying or dividing are known as derived quantities.

Quantities

② These are only 7 in numbers.

③ These are large in number.

Units

③ Units that can be used for base quantities are known as base units.

③ Units that can be used for derived quantities are known as derived quantities.

Examples

- ① Length / meter / m
- ② Mass / kilogram / kg
- ③ time / seconds / s

- ① velocity : $\text{m/s} = \text{m} \div \text{s}$
- ② force : $\text{N} = \text{kg} \cdot \text{m/s}^2$
- ③ acceleration : $\text{a} = \text{m/s}^2$

Q.2(ii)

A.2(ii)

Distance :-

The longest path traveled b/w two positions is known as distance. Distance has no direction so it is a scalar quantity. Distance is usually denoted by Δx , Δr , Δs , Δd or ΔL and its SI unit is meter (m).
formula :- $\Delta d = d_f - d_i$

OR

$$d = \text{Average} \times t.$$

Displacement :-

Shortest distance travelled from initial position to final position is called displacement. Displacement has direction so it is a vector quantity. Displacement SI unit is same as distance (meter : m).
formula :-

$$\vec{S} = \vec{v} \times t$$

$$\Delta \vec{d} = \vec{d}_f - \vec{d}_i$$

② i. Distance :-

3km north the 4km east :- ~~3+4~~ $3+4 = 7\text{km}$.

ii. Displacement :-

$$= \sqrt{(3)^2 + (4)^2}$$

$$= \sqrt{9+16} \Rightarrow \sqrt{25} \Rightarrow 5\text{km (at an angle NE)}$$

Q: 2(iii)
A: 2(iii)

Newton's second law:-

Acceleration of an object is directly proportional to the magnitude of a net force, in a same direction, is inversely proportional to the mass of an object.

Formula derived is

$$a \propto F$$

$$a \propto \frac{1}{\text{mass}}$$

$$a \propto \frac{F}{m} \Rightarrow a = \frac{F}{m}$$

$$\Rightarrow F = ma$$

① Acceleration :- $\frac{F}{m}$

~~60N~~ $\frac{60N}{12Kg}$

$$= 5m/s^2$$

Unit of force :- This statement also enables us to form system international (SI) unit of force that is Newton and it is represented by symbol "N". $F = ma$
 $F = Kg \cdot m/s^2$

Q.2(iv)
A.2(iv)

Statement :-

When one object exerts force on a second object, the second object exerts an equal and opposite force on the first object.

Explanation :-

When an object 'A' exerts force on an object 'B' written as ' $\vec{F}_{AB}$ ', object 'B' also exerts an equal force on object 'A' written as ' $\vec{F}_{BA}$ ', but in different opposite :-

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

Example :-

⇒ When we jump, our legs apply a force to the ground, and the ground applies an equal and opposite reaction force that pushes us into the air.

⇒ The exhaust from the rocket creates a downward force which creates an equal and opposite thrust in the upward direction.

Q: 2(v)

A: 2(v)

Torque $\tau = F \cdot d$, the turning effect produced by a force about a fixed point, where displacement is perpendicular distance (moment arm).

It's SI unit is N.m.

Principle of moments:

for an object in equilibrium the sum of clockwise moments must be equal to the sum of anticlockwise moments about any pivot.

$$\text{Torque} = 40\text{ N} \times 0.25\text{ m} = 10\text{ Nm}.$$

Q: 2(vi)(or)

A: 2(vi)(or)

Friction :-

A contact force that resists relative motion b/w surfaces in contact, acts opposite to motion.

SI unit is newton (N). It's type are sliding, rolling, and fluid surface.

Advantages:

Enables walking e.g. (friction b/w shoes and ground) e.g. car brakes work through friction.

Disadvantages:-

Causes ~~water~~ wear and tear of machine parts.

Q: 2(vii)
A: 2(vii)

Hooke's law:-

Within the elastic limit, the extension x of a spring constant is directly proportional to applied force F . Mathematically,

$$F = Kx \quad (K = \text{spring constant}).$$

K = restoring force / unit extension.
SI unit:- N/m

$$K = \frac{F}{x} \Rightarrow \frac{20N}{0.05m} = 400N/m.$$

Q: 2(viii) or
A: 2(viii) (or)

$$m = \rho Ah, \quad f = \rho Ahg$$

$$\rho = \frac{F}{A}$$

$$\frac{\rho Ahg}{A}$$
$$\rho \cdot gh$$

Q: 2 (in)

A: 2 (in)

Work done: $W = Fd$. SI unit: joule = N.m.

$$W = 150 \text{ N} \times 5 \text{ m} = 750 \text{ J.}$$

Work done is zero when force is perpendicular to displacement.

Q: 2 (n)

A: 2 (n)

Power: $P = \frac{W}{t}$. SI unit: Watt (W)

Efficiency: $\eta = \left(\frac{\text{Useful output energy}}{\text{input energy}} \right) \times 100\%$

a) Efficiency: $\frac{4800}{6000} \times 100\% = 80\%$

b) Useful power output: $\frac{4800}{20} = 240 \text{ W}$

Q: 2 (ni)

A: 2 (ni)

Hypothesis:

A tentative assumption or explanation made before any research, which can be tested and

verified.

Theory :-

A well-supported description of natural phenomena verified through rigorous testing ; explains how nature behaves.

Law :- A statement of an observed regularity giving the relationship b/w variables ; describe what happens but not always why.

Section- C

Q: 3(i)(or) _____
A: 3(i)(or) _____

Translatory motion :-

every part of the body moves in the same direction.

Types: rectilinear (straight line, e.g. car on a straight road), curvilinear (curved path) & random.

Rotatory motion :-

Body spins about an axis (e.g. Earth rotating on its axis & fan blades).

Vibratory motion :- oscillation about a fixed point.

Distance:

A path travelled b/w two positioned ~~area~~ call points known as distance. Distance is a scalar quantity.

Displacement:

Two shortest path b/w initial position to final position.

Speed:

Rate of change of distance
 Δt is a scalar quantity.

Velocity:

Rate of change of displacement
 Δt is a vector quantity.

Q: 3(ii)

A: 3(ii)

First Law:-

If net internal force on an object is zero, it will remain at rest or continue in uniform straight line motion.

$$F_{\text{net}} = 0 \Rightarrow \Delta v = 0$$

Example:

Passenger jerk forward when a bus suddenly brakes

Second law :- $F_{\text{net}} = ma$. The acceleration of an object is directly proportional to the magnitude of the net force in a same direction, inversely proportional to the mass of an object.

Third law :-

$F_{AB} = -F_{BA}$. For every action there is an equal and opposite reaction acting on a different body.

Q: 3 (iii) (or)

A: 3 (iii) (or)

Elasticity :-

The ability of a deformed body to return to its original shape and size when the deforming force is removed. Materials that do not recover is called elastic limit.

Hooke's law :-

Within the elastic limit, the extension x is directly proportional to the applied force F .

$$F = kx \quad (k = \text{spring constant}).$$

Force - extension graph :-

Straight line through origin up to point A (limit of proportionality),

point B. (elastic limit) (beyond which permanent deformation occurs.
 point C :- (ultimate stress) point D
 (breaking point)

$$Q: 3 \text{ (iv) (or)}$$

$$A: 3 \text{ (iv) (or)}$$

$$\text{Power} = P = \frac{W}{t} = \frac{E}{t} \quad \text{SI unit} = \text{Watt.}$$

($W = J/s$) It measures how quickly energy is converted. Efficiency $\eta =$
 $\left(\frac{\text{useful energy output}}{\text{input energy}} \right)$

$$(a) \eta = \frac{E_{\text{out}}}{E_{\text{input}}} \times 100\%$$

$$\frac{120}{4.00} \times 100\% = 30\%$$

$$(b) \text{ Useful power} = \frac{E_{\text{out}}}{t} = \frac{120}{0.5} = 240 \text{ W}$$

A machine can never be 100% efficient because frictional forces always convert some input energy into waste thermal energy. This waste heat is not useful for the intended purpose of the machine.

MCGs:

1. B
2. B
3. C
4. B
5. B
6. B
7. C
8. C
9. C
10. C
11. B
12. D